Fiscal Year:	FY 2018	**Task Last Updated:**	FY 02/14/2018
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Project Title:	High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity		
Division Name:	Physical Sciences		
Program/Discipline:			
Program/Discipline--Element/Subdiscipline:	FUNDAMENTAL PHYSICS--Fundamental physics		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2013 Fundamental Physics NNH13ZTT002N (Cold Atom Laboratory--CAL)
Start Date:	04/01/2014	**End Date:**	04/30/2019
No. of Post Docs:	0	**No. of PhD Degrees:**	1
No. of PhD Candidates:	4	**No. of Master' Degrees:**	0
No. of Master's Candidates:	0	**No. of Bachelor's Degrees:**	0
No. of Bachelor's Candidates:	0	**Monitoring Center:**	NASA JPL
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Flight Program:	ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:	Feb 2018 report: No changes.		
COI Name (Institution):	Malinovsky, Vladimir Ph.D. (Stevens Institute of Technology)		
Grant/Contract No.:	JPL 1501691		
Performance Goal No.:			
Performance Goal Text:			

Objective: The system of fundamental constants (the speed of light, Planck's constant, the fine structure constant, the Rydberg constant, etc.) is important because it directly reflects our current understanding of the foundations of nature. The objective of the project is to measure one of these constants, the Rydberg constant, at an uncertainty level in the range of a few parts in 10^12 (the current uncertainty stands at 5.9 parts in 10^12). It is planned to use circular Rydberg atoms, which are very large, highly excited atoms with doughnut-shaped electron distributions that do not penetrate into the atomic nucleus. As a result, these atoms are not susceptible to perturbations caused by electron wavefunction overlap with the nuclear charge density and quantum-electrodynamics (QED) corrections. The measurement therefore yields a result for the Rydberg constant that is not contingent to the accuracy/precision of input from other areas of physics. It is this property that makes a measurement of the Rydberg constant with circular Rydberg atoms particularly useful, as it

Task Description:	will present a valuable complement/check for results based on established methods. Due to the absence of critical QED corrections and nuclear penetration, a measurement of the Rydberg constant with circular Rydberg atoms would help resolve a recently reported inconsistency, where it was concluded that "... either the Rydberg constant has to be shifted by -110kHz/c (4.9 standard deviations), or the calculations of the QED [quantum-electro-dynamic] effects in atomic hydrogen or muonic hydrogen atoms are insufficient." [R. Pohl et al., Nature 466, 213 (2010)]. Method: Rb atoms are cooled to below 1 microK and laser-excited into a low-angular-momentum Rydberg level with principal quantum number n on the order of 40. Using an adiabatic atomic-state transformation method, the Rydberg atoms are then transferred into the circular state |n,n1=0,n2=0,m=n-1> (m, n1, and n2 are the magnetic and parabolic quantum numbers). An optical lattice (wavelength 1064 nm or 532 nm on the ground, 850 nm at CAL) is adiabatically ramped on. The Rydberg atoms are trapped in the lattice via the ponderomotive interaction of the Rydberg electron with the lattice light. During the hold time of the atom cloud in the lattice, the electric-quadrupole transition into the state |n+2,n1=1,n2=1,m=n-1> is driven by amplitude-modulating the optical lattice at the transition frequency. The transition is probed using state-selective electric-field ionization. The measured transition frequency allows one to extract the Rydberg constant. Key advantages of the method are that the transition is free of first-order Stark and Zeeman shifts, that QED, nuclear-penetration, hyperfine, and other shifts are small, and that the optical lattice causes only moderate trap-induced transition shifts. Theoretical work includes the modeling of the circular-state preparation using multi-level adiabatic passage and of the circular-state transition dynamics in a shallow, microwave-modulated optical lattice, in which center-of-mass wave-packet expansion and tunneling must be considered. Relevance to NASA and relevance of microgravity: High-precision measurement of fundamental constants represents an important part of contemporary atomic-physics research. The proposed high-precision measurement of the Rydberg constant adds a new component to the NASA Fundamental Physics program, and to the Cold Atom Laboratory (CAL) research portfolio. To achieve optimal spectroscopic performance it is critical to reduce systematic lattice-induced transition shifts and related shifts. Since the most critical shifts diminish with decreasing optical-lattice depth, the lattices must be chosen as shallow as possible. Microgravity conditions are important because they enable the use of shallow lattices without the atoms falling out. The microgravity environment provided by the CAL/ its successor instrument, including its 850 nm optical-lattice laser, offers an ideal platform for the project to reach its ultimate uncertainty goal. An International Space Station (ISS)-based setup could also provide a sufficient period of measurement time to acquire good statistics in the high-precision spectra. More generally, the research could bring Rydberg atoms into space. This has far-reaching additional potential for the development of novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of the physics of very dilute, but strongly interacting Bose Einstein Condensates.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	High-precision spectroscopy and metrology impacts fields such as time measurement and navigation. Benefits to the general public include atomic clocks and the GPS system. The project “High-Precision Microwave Spectroscopy of Long-Lived Circular-State Rydberg Atoms in Microgravity” involves the development of a fundamentally new type of spectroscopy, which has been successfully demonstrated during the first year of the grant. The future development of laser systems to hone this innovative technique may lead to benefits in other disciplines, such as chemistry, biology, and engineering. The research involves graduate students, undergraduate students, and post-docs, who are being prepared for future jobs in academia, government, and industry. Having a capable scientific workforce is essential for progress and competitiveness in science and technology, as well as for the national economy. The main goal of the project, which is to measure the Rydberg constant with circular Rydberg atoms, will help resolve the “proton radius puzzle.” The puzzle has arisen from inconsistencies in measurements on low-lying electronic and muonic states of hydrogen, which involve the electromagnetic interaction, QED, the proton structure, and QCD. The project will therefore serve as an indirect test for QED and the structure of the proton. With Rydberg atoms, the Rydberg constant will be tested at electron energies in the range of milli-electronvolts. This regime is rather different from the energy regime of previous measurements (eVs to keVs). More generally, compact two-frequency laser sources can bring Rydberg atoms into space, which carries the potential for novel quantum technologies for microwave / THz / thermal-field sensing and imaging, and well as for exploration of new phases of matter in very dilute, but strongly interacting, Rydberg-dressed Bose Einstein Condensates.
	A measurement of the Rydberg constant using cold circular Rydberg atoms is proposed. Since these atoms have long lifetimes, negligible quantum electrodynamics and no nuclear overlap level shifts, the measurement can shed new light on the “proton radius puzzle.” The atoms are trapped in an optical lattice. Transitions are driven using a lattice-modulation technique to perform Doppler-free sub-THz spectroscopy; the transition frequencies yield the Rydberg constant. The selected transitions are free of first-order Zeeman and Stark shifts. Systematic uncertainties due to lattice-induced shifts, core-polarization, second order Zeeman and Stark shifts, etc. must all be estimated and controlled. The measurement will be most accurate and precise in the shallow-lattice, low-temperature environment afforded by microgravity. A quantum model of transitions driven by the lattice-modulation technique must be developed to accurately model spectroscopic lines in this quantum regime. Progress: (1) Comprehensive study of the systematics in the pursued high-precision measurement. Understanding the systematics is of utmost importance in any high-precision measurement (HPM). The team has been working on effort since several years. There was no rush in getting this published. It was far more important that the team was convinced the results are correct, that the list of systematics is complete, and that all systematics are thoroughly understood and well modeled. The paper was published in 2017, “Measuring the Rydberg constant using circular Rydberg atoms in an intensity-modulated optical lattice,” Andira Ramos, Kaitlin Moore, and Georg Raithel, PHYSICAL REVIEW A 96, 032513 (2017) and arXiv:1705.02682 [physics.atom-ph]. The Principal Investigator (PI) hopes that there may be useful feedback and possibly additional insights coming back from the community. Progress (2). Quantum-defect and core polarizability measurements. One of the systematics is associated with the ionic core polarizability of rubidium. The team is working on this and has submitted a paper titled “Measurement of Rb G-series quantum defect using two-photon nG to (n+2)G microwave spectroscopy.” Two-photon high-precision spectroscopy is engaged to precisely measure the high-angular-momentum G-series quantum defect of Rb-85. The transition frequencies are in the sub-THz regime. Measurements of atomic transition frequencies are the cornerstone of precision spectroscopy, used in applications

Task Progress:	ranging from atomic clocks to measuring gravitational redshifts and the radius of the proton. Often, cold atoms are used in these measurements. Alkali atoms, which have a single valence electron similar to hydrogen, are easier to laser-cool than hydrogen due to a lower recoil energy and near-infrared cooling-transition wavelengths. However, in an alkali atom such as rubidium (Rb), the interaction between the ionic core of the atom and the valence electron depresses the energy levels of the valence electron below the expected hydrogenic levels (the "quantum defect"). In Rydberg-atom-based high precision spectroscopy, it is imperative to precisely quantify this quantum defect for the commonly-used alkali species. For electrons in high-angular-momentum states, the quantum defect is dominated by the polarizability of the ionic core, which may be extracted from high-angular-momentum defect measurements. Since in the current NASA-funded project all of the above applies, the team pursues HPM of rubidium quantum defects and dipolar core polarizabilities (the quadrupolar polarizability term in not important for circular-state Rydberg atoms). In the most recent previous experimental measurement of the nG-series quantum defect, microwave spectroscopy of nD to (n+1)G [J. Lee, J. Nunkaew, and T. Gallagher, Phys. Rev. A 94 022505 (2016)] was performed, whereas in the submitted work sub-THz spectroscopy was used to measure nG to (n+2)G transitions in a field-free environment. Furthermore, the presently used two-photon transition depends only on one set of quantum defects and takes advantage of equal Lande g-factors in the lower and upper states, which is a great advantage. In the present experiment, the samples of cold nG Rydberg atoms were prepared via a three-photon optical excitation and controlled electric-field mixing and probed with 40-microsecond long microwave interaction pulses. Results were compared with other recent measurements, performed elsewhere, and the new (our) measurement were found to be consistent and one order of magnitude more precise. The experimental procedure included a careful cancellation of stray electric fields in all three dimensions. In the submitted paper, future extensions towards precision measurements of atomic polarizabilities are discussed. Progress (3) Modeling of lattice modulation spectroscopy in the ultracold regime. The experimental work is accompanied by a theoretical effort lead by the co-PI V. Malinovsky, with some associated modeling work provided by the PI. The present objective of the theory work is to model the spectroscopic line shapes of Rydberg transitions in an amplitude-modulated Rydberg-atom lattice in a manner that takes the quantization of the center-of-mass motion into account. This ability is an important ingredient to determine the line center with 1Hz uncertainty, as required in the present project. To take into account the quantization of motion of the Rydberg atoms in the optical lattice, a theoretical model has been developed in which both ground and excited wave functions are subject to the periodic potentials, representing the lattice fields, while the coupling (the effective Rabi frequency) is also periodic as function of the translational coordinate. To obtain the spectrum of the excited-state population, the time-dependent Schroedinger equation has been solved in the momentum representation. The spectra are also modeled using Fermi's Golden Rule on transitions between thermally populated spinor Bloch states in the optical lattice. The results are all in place. Current work is focused on finalizing two manuscripts on the theory work. The emphases are to (a) make it very clear why this is a new method of Doppler-free spectroscopy, (b) emphasize the utility and the limits when applied in very shallow lattices afforded under microgravity conditions, and (c) to discuss the portability to other spectral regions and molecular-size domains (such as molecules in x-ray lattices that are modulated at optical frequencies). Progress (4) Experimental tasks. In the last year, the team has installed a 1260-nm laser amplifier from Thorlabs and procured two laser systems from Moglabs (a 776-nm and a 1260-nm master laser). The 776-nm system has already been delivered and is being tested. The lasers have a specified linewidth of 200 kHz, when locked using the built-in Pound-Drever-Hall lock (which feeds back on an internal laser etalon and the laser current). The senior student (Kaitlin Moore) has graduated, and a new student is being trained (Ryan Cardman). To minimize the effect of the student turnover on the progress of the project, I have assigned a second senior student (half-time) to the project. The current spectroscopic work is focused on electric-field zeroing and Rydberg-atom circularization. As a start into circular-state spectroscopy work, the team has procured a microwave frequency doubler, waveguides and a microwave horn for two-photon microwave spectroscopy of circular-state transitions of the type n to n+1 (off-resonant) to n+2. There also is continued sub-THz spectroscopy work on the nH-series quantum defect. Progress (5). Conference submissions. The work was presented at national and international conferences and workshops (last at the Atomic Physics workshop in Dresden, Germany). The work was presented at Division of Atomic, Molecular and Optical Physics (DAMOP) conference 2017 (K1.00082 Blackbody effects in high-precision microwave spectroscopy with circular Rydberg atoms, and Q1.00024 Progress towards measuring the Rydberg constant with circular Rydberg atoms), and will also be presented at DAMOP 2018, and the NASA research conference in La Jolla. The PI has submitted a two-page paper abstract to the CPEM 2018 (Conference on precision electromagnetic measurements). This conference will have a session focused on fundamental constants and the proton radius. The accepted conference papers will be published on Institute of Electrical and Electronics Engineers (IEEE).

Bibliography Type:	Description: (Last Updated: 02/24/2025)
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Articles in Peer-reviewed Journals	Ramos A, Moore K, Raithel G. "Measuring the Rydberg constant using circular Rydberg atoms in an intensity-modulated optical lattice." Physical Review A. 2017;96:032513. Published 29 September 2017. https://doi.org/10.1103/PhysRevA.96.032513 , Sep-2017
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